FREE/OPEN SOURCE SOFTWARE QUALITY:
EXPLORING THE IMPACT OF DEFECT MANAGEMENT

Summary

Free/Open Source Software (F/OSS) is an evolving paradigm of software development which allows the entire Internet community to use its combined programming knowledge, creativity and expertise to develop software solutions, which could render benefits to whole community without involving cost and proprietary issues. F/OSS Participants rely on extensive peer collaboration through the Internet using Version Control System, Mailing List, Defect Management System, Internet Relay Chat, Discussion Forum etc. These tools enable participants to collaborate in the F/OSS development process as well as act as repositories to store the communication activities of the participants.

Even though there are number of qualitative and quantitative studies about F/OSS, little attention has been paid to the rich information stored in Defect Management Systems of F/OSS Projects. Moreover, little functionality exists to aid the users in evaluating and selecting the most appropriate software from the thousands of F/OSS Projects available. There is a considerable requirement of efforts in Defect Management specifically focusing upon analyzing the past experience, degree of improvement or deterioration and responsiveness towards users for various F/OSS Projects.

To gain an insight on current status in Defect Management among various F/OSS Projects, a survey has been conducted among F/OSS developers using online questionnaire. A total of 177 responses received from 137 projects are used to quantify certain attributes like general characteristics of F/OSS Projects, use of Defect Management Systems, the perception of F/OSS developers about effectiveness of defect reporting and efficiency of defect resolution. The findings show that majority of the projects use Defect Management System, but the level of consistency and completeness is found to be higher in case of larger projects. The quantitative evidence shows quite high participation of user community in form of identifying bugs as compared to requesting features and

submitting patches towards F/OSS Projects. The study establishes that the Defect Management needs to be explored further by using publicly accessible defect repository to provide useful feedback on F/OSS Projects.

To obtain detailed defect data on F/OSS Projects, access is obtained from a research collaboratory i.e. SourceForge Research Data Archive (SRDA) which maintains the monthly database dumps in form of multiple-layer data schema and supports web based interface to deliver the formatted information. As F/OSS Projects have large number of defect reports, Defect Analysis and Reporting Tool (DART) is developed using F/OSS technologies to process the defect data based on several parameters and to generate reports in textual/graphical formats.

To determine the impact of Defect Management on F/OSS Quality, 20 F/OSS Projects are selected from diverse application domains. The defect data is analyzed from various perspectives, with special focus on evaluating the efficiency and effectiveness in resolving defects and determining responsiveness towards users. During the analyses, it has been found that generally defect resolution is not performed consistently which results in declining defect removal rate and an ever increasing average age of defect resolution. Backlog of defects increases gradually but reduces in bursty manner. It is also found that a few defects remain pending for fairly long period of time and hence cause increasing trend in overall defect pending age. Defect reporting is found to be a widely dispersed type of action mostly contributed by users external to core team while defect resolution is concentrated among a few individuals mainly from core team. Generally, the most active users in the projects carry out most of the tasks while others contribute only once or twice.

Finally, several problems and inefficiencies have been identified in Defect Management which may have serious effects on F/OSS Quality in the long term if effective countermeasures are not taken. The effective ways and means are suggested as well as a process is proposed to distribute roles and responsibilities among F/OSS participants which can help F/OSS Projects to improve the effectiveness and efficiency of Defect Management and hence assure better quality of F/OSS Projects.